

Statistics for Insurance — Chapter 1

Worked Solutions to Exercises 1–8 (*Section 1.8*)

fully detailed — every calculation written out

Study notes: *problem • setup • tools used • full solution*

For each exercise: the statement, the setup, the tools invoked, and a complete proof in which every algebraic step is shown — including where each substitution and expression comes from. A short **Technique** note closes each item.

Legend: ■ Problem ■ Setup ■ Tools used ■ Solution ■ Technique

1. Two-sided Bernstein inequality

Problem

By applying Theorem 1.3 (Bernstein) to $-\gamma$, show that $\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n \gamma(Z_i)\right| \geq \kappa t + \sqrt{2t}\right) \leq 2e^{-nt}$, and deduce a constant $C > 0$ such that for all $\varepsilon > 0$, $\mathbb{P}(|\bar{\gamma}_n| \geq \varepsilon) \leq 2 \exp\left(-\frac{Cn\varepsilon^2}{1+\kappa\varepsilon}\right)$, where $\bar{\gamma}_n = \frac{1}{n} \sum_{i=1}^n \gamma(Z_i)$.

■ **Setup** (Z_n) are independent, $\mathbb{E}[\gamma(Z_n)] = 0$, and the moment bound $\frac{1}{n} \sum_{i=1}^n \mathbb{E}|\gamma(Z_i)|^m \leq \frac{m!}{2} \kappa^{m-2}$ holds for all $m \geq 2$. Theorem 1.3 delivers the *one-sided* bound

$$\mathbb{P}(\bar{\gamma}_n \geq \kappa t + \sqrt{2t}) \leq e^{-nt}, \quad t > 0. \quad (\star)$$

We must (i) turn $(\star)$ into a two-sided bound, and (ii) rewrite the threshold $\kappa t + \sqrt{2t}$ as an explicit rate in the level ε .

■ **Tools used** Bernstein's inequality (Thm 1.3); the union bound $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$; the algebraic inversion of $\varepsilon = \kappa t + \sqrt{2t}$.

■ Solution

Step 1 (symmetrise). Apply Theorem 1.3 to the function $-\gamma$. Its hypotheses hold with the *same* κ , because

$$\mathbb{E}[-\gamma(Z_n)] = -\mathbb{E}[\gamma(Z_n)] = 0, \quad \frac{1}{n} \sum_{i=1}^n \mathbb{E}|-\gamma(Z_i)|^m = \frac{1}{n} \sum_{i=1}^n \mathbb{E}|\gamma(Z_i)|^m \leq \frac{m!}{2} \kappa^{m-2}.$$

Since $\frac{1}{n} \sum_i (-\gamma(Z_i)) = -\bar{\gamma}_n$, the conclusion $(\star)$ applied to $-\gamma$ reads

$$\mathbb{P}(-\bar{\gamma}_n \geq \kappa t + \sqrt{2t}) \leq e^{-nt}, \quad \text{i.e.} \quad \mathbb{P}(\bar{\gamma}_n \leq -(\kappa t + \sqrt{2t})) \leq e^{-nt}.$$

Writing $a := \kappa t + \sqrt{2t}$ and using $\{|\bar{\gamma}_n| \geq a\} = \{\bar{\gamma}_n \geq a\} \cup \{\bar{\gamma}_n \leq -a\}$, the union bound gives

$$\mathbb{P}(|\bar{\gamma}_n| \geq \kappa t + \sqrt{2t}) \leq \mathbb{P}(\bar{\gamma}_n \geq a) + \mathbb{P}(\bar{\gamma}_n \leq -a) \leq e^{-nt} + e^{-nt} = 2e^{-nt}. \quad (1)$$

Step 2 (reparametrise level $\leftrightarrow t$). We want a bound at a *given* level $\varepsilon > 0$, but (1) is stated at the level $\kappa t + \sqrt{2t}$. So we *choose* t to make these levels equal:

$$\kappa t + \sqrt{2t} = \varepsilon. \quad (2)$$

The left side is continuous and strictly increasing in t , running from 0 (at $t = 0$) to ∞ , so (2) has a unique solution $t = t(\varepsilon) > 0$. Substituting into (1),

$$\mathbb{P}(|\tilde{\gamma}_n| \geq \varepsilon) \leq 2e^{-nt}. \quad (3)$$

It remains only to bound t from below by something of the form $\frac{C\varepsilon^2}{1+\kappa\varepsilon}$, so that $e^{-nt} \leq e^{-Cn\varepsilon^2/(1+\kappa\varepsilon)}$.

Step 3 (invert the rate — this is where the $\varepsilon^2/(1+\kappa\varepsilon)$ comes from). We seek $C > 0$ with

$$t \geq \frac{C\varepsilon^2}{1+\kappa\varepsilon} \iff R := \frac{t(1+\kappa\varepsilon)}{\varepsilon^2} \geq C.$$

So the whole task is to compute the ratio R and bound it below. To evaluate R we must express t and ε through *one* variable, using their link (2). The clean choice is

$$\boxed{s := \sqrt{2t} \geq 0} \implies t = \frac{s^2}{2}, \quad \sqrt{2t} = s.$$

Then (2) becomes

$$\varepsilon = \kappa t + \sqrt{2t} = \kappa \frac{s^2}{2} + s = s \left(1 + \frac{\kappa s}{2}\right).$$

Introduce a second shorthand $\boxed{x := \kappa s \geq 0}$, so that $\varepsilon = s(1 + \frac{x}{2})$. Now compute each ingredient of R separately:

$$\begin{aligned} \varepsilon^2 &= s^2 \left(1 + \frac{x}{2}\right)^2, \\ \kappa\varepsilon &= \kappa s \left(1 + \frac{x}{2}\right) = x \left(1 + \frac{x}{2}\right) = x + \frac{x^2}{2} \implies 1 + \kappa\varepsilon = 1 + x + \frac{x^2}{2}, \\ t(1 + \kappa\varepsilon) &= \frac{s^2}{2} \left(1 + x + \frac{x^2}{2}\right). \end{aligned}$$

Divide the last by ε^2 ; the factor s^2 cancels:

$$R = \frac{t(1 + \kappa\varepsilon)}{\varepsilon^2} = \frac{\frac{s^2}{2} \left(1 + x + \frac{x^2}{2}\right)}{s^2 \left(1 + \frac{x}{2}\right)^2} = \frac{\frac{1}{2} \left(1 + x + \frac{x^2}{2}\right)}{\left(1 + \frac{x}{2}\right)^2}.$$

Expand the denominator, $\left(1 + \frac{x}{2}\right)^2 = 1 + x + \frac{x^2}{4}$, so

$$R = \frac{\frac{1}{2} \left(1 + x + \frac{x^2}{2}\right)}{1 + x + \frac{x^2}{4}}.$$

Finally we bound $R \geq \frac{1}{2}$. Multiplying out,

$$R \geq \frac{1}{2} \iff \frac{1}{2} \left(1 + x + \frac{x^2}{2}\right) \geq \frac{1}{2} \left(1 + x + \frac{x^2}{4}\right) \iff \frac{x^2}{2} \geq \frac{x^2}{4},$$

which holds for every $x \geq 0$. Hence $R \geq \frac{1}{2}$, i.e.

$$t \geq \frac{\varepsilon^2}{2(1 + \kappa\varepsilon)}.$$

Conclusion. Insert this lower bound for t into (3): since $x \mapsto e^{-nx}$ is decreasing,

$$\mathbb{P}(|\tilde{\gamma}_n| \geq \varepsilon) \leq 2e^{-nt} \leq 2 \exp\left(-\frac{n\varepsilon^2}{2(1 + \kappa\varepsilon)}\right), \quad \text{so } \boxed{C = \frac{1}{2}}. \quad \blacksquare$$

Sanity check of the two regimes. For small ε (small x), $R \rightarrow \frac{1}{2}$ and the exponent is $\approx -\frac{1}{2}n\varepsilon^2$ (sub-Gaussian). For large ε (large x), $R \rightarrow \frac{\frac{1}{2}x^2/2}{x^2/4} = 1$ and, since $1 + \kappa\varepsilon \approx \kappa\varepsilon$, the exponent is $\approx -\frac{n\varepsilon}{2\kappa}$ (sub-exponential). This is the expected Bernstein shape.

■ **Technique** Symmetrise a one-sided tail bound by re-applying it to $-\gamma$; convert a *threshold*-form bound into a *rate*-form bound by solving $\varepsilon = \kappa t + \sqrt{2t}$. The trick that makes the algebra collapse is the single-variable substitution $s = \sqrt{2t}$ then $x = \kappa s$: the awkward $\sqrt{2t}$ disappears and s^2 cancels, leaving a rational function of x bounded below by $\frac{1}{2}$.

2. Strong law fails when $\mathbb{E}|Z_1| = \infty$

Problem

Let (Z_n) be iid with $\mathbb{P}(Z_1 > z) = z^{-\alpha}$ for $z \geq 1$, $\alpha \in (0, 1)$. **(a)** Show $\mathbb{E}|Z_1| = \infty$. **(b)** Show $\limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n Z_i = \infty$, so Theorem 1.2 fails without the moment assumption. *Hint: Borel–Cantelli with $\{Z_n > Mn\}$.*

■ **Setup** $Z_1 \geq 1$ a.s. (support $[1, \infty)$), heavy Pareto tail with index $\alpha < 1$. Put $S_n = \frac{1}{n} \sum_{i=1}^n Z_i$. We build events capturing a single anomalously large summand and transfer them to S_n .

■ **Tools used** Tail-integral identity $\mathbb{E}Y = \int_0^\infty \mathbb{P}(Y > z) dz$ (valid for $Y \geq 0$); divergence of the p -series with $p = \alpha < 1$; the second Borel–Cantelli lemma (Lemma 1.1, independence); positivity ($S_n \geq Z_n/n$).

Solution

(a) Because $Z_1 \geq 1 > 0$ we may use the tail-integral identity. Note $\mathbb{P}(Z_1 > z) = 1$ for $z < 1$ (as $Z_1 \geq 1$) and $\mathbb{P}(Z_1 > z) = z^{-\alpha}$ for $z \geq 1$, hence

$$\mathbb{E}Z_1 = \int_0^\infty \mathbb{P}(Z_1 > z) dz = \underbrace{\int_0^1 1 dz}_{=1} + \int_1^\infty z^{-\alpha} dz.$$

Evaluate the second integral explicitly (its antiderivative is $\frac{z^{1-\alpha}}{1-\alpha}$ since $\alpha \neq 1$):

$$\int_1^\infty z^{-\alpha} dz = \lim_{b \rightarrow \infty} \left[\frac{z^{1-\alpha}}{1-\alpha} \right]_1^b = \lim_{b \rightarrow \infty} \frac{b^{1-\alpha} - 1}{1-\alpha} = +\infty,$$

because $1 - \alpha > 0$ forces $b^{1-\alpha} \rightarrow \infty$. Therefore $\mathbb{E}Z_1 = \infty$, and $\mathbb{E}|Z_1| = \mathbb{E}Z_1 = \infty$.

(b) Fix $M > 0$ and set $A_n = \{Z_n > Mn\}$.

Independence and divergence. Each A_n depends only on Z_n , and the Z_n are independent, so the A_n are independent. For $n \geq 1/M$ (so that $Mn \geq 1$ and the tail formula applies),

$$\mathbb{P}(A_n) = \mathbb{P}(Z_n > Mn) = (Mn)^{-\alpha} = M^{-\alpha} n^{-\alpha}.$$

Summing,

$$\sum_{n \geq 1/M} \mathbb{P}(A_n) = M^{-\alpha} \sum_{n \geq 1/M} n^{-\alpha} = \infty,$$

since $\sum_n n^{-\alpha}$ is a p -series with $p = \alpha < 1$, hence divergent. By the second Borel–Cantelli lemma (independence), $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Transfer to the average. All summands are $\geq 1 > 0$, so dropping every term except the last,

$$\sum_{i=1}^n Z_i \geq Z_n \quad \implies \quad S_n = \frac{1}{n} \sum_{i=1}^n Z_i \geq \frac{Z_n}{n}.$$

On the event A_n we have $Z_n > Mn$, hence $Z_n/n > M$, hence $S_n > M$. Since A_n occurs for infinitely many n (a.s.), $S_n > M$ for infinitely many n , and therefore

$$\limsup_{n \rightarrow \infty} S_n \geq M \quad \text{a.s.}$$

From “every M ” to ∞ . Apply the above with $M = k$ for each $k \in \mathbb{N}$; let $N_k = \{\limsup_n S_n < k\}$, a null set. Then $N = \bigcup_{k \in \mathbb{N}} N_k$ is null, and off N we have $\limsup_n S_n \geq k$ for every k , i.e. $\limsup_n S_n = \infty$ a.s. In particular S_n has no finite limit, so the SLLN (Thm 1.2) fails without integrability. ■

■ **Technique** “One big term” via the *second* Borel–Cantelli lemma. Heavy tails ($\alpha < 1$) make $\sum \mathbb{P}(Z_n > Mn)$ diverge, so a summand of order $\gtrsim Mn$ recurs forever; positivity ($S_n \geq Z_n/n$) transfers this to the average. Keep $\mathbb{E}Y = \int_0^\infty \mathbb{P}(Y > z) dz$ ready.

3. CLT fails when $\mathbb{E}[Z_1^2] = \infty$

Problem

Let (Z_n) be iid *symmetric* with $\mathbb{P}(|Z_1| > z) = z^{-\alpha}$ for $z \geq 1$, $\alpha \in (1, 2)$. Show the Lindeberg condition for the normalisation $\sqrt{n}$ fails by proving $n \mathbb{E}[Z_1^2; |Z_1| > \varepsilon\sqrt{n}] \rightarrow \infty$ for every $\varepsilon > 0$; show $n^{-1/2} \sum_{i=1}^n Z_i$ cannot converge to a non-degenerate normal law, so Theorem 1.6 fails.

■ **Setup** From $\mathbb{P}(|Z_1| > z) = z^{-\alpha}$ the cdf of $|Z_1|$ is $1 - z^{-\alpha}$ on $[1, \infty)$, so $|Z_1|$ has density $f_{|Z_1|}(z) = \alpha z^{-\alpha-1}$, $z \geq 1$. Put $S_n = n^{-1/2} \sum_{i=1}^n Z_i$ and $M_n = \max_{i \leq n} |Z_i|$.

■ **Tools used** Direct moment/tail integrals; the elementary bound $Z_1^2 > a^2$ on $\{|Z_1| > a\}$; $1 - x \leq e^{-x}$; the tail necessary condition $n\mathbb{P}(|Z_1| > \varepsilon\sqrt{n}) \rightarrow 0$ for a Gaussian limit; Lévy’s maximal inequality for symmetric variables.

■ Solution

Moments (why the variance is infinite). Using the density,

$$\mathbb{E}[Z_1^2] = \int_1^\infty z^2 \alpha z^{-\alpha-1} dz = \alpha \int_1^\infty z^{1-\alpha} dz = \alpha \lim_{b \rightarrow \infty} \frac{b^{2-\alpha} - 1}{2 - \alpha} = +\infty,$$

because $2 - \alpha > 0$ (as $\alpha < 2$) makes $b^{2-\alpha} \rightarrow \infty$. By symmetry $\mathbb{E}Z_1 = 0$, while $\mathbb{E}|Z_1| = \alpha \int_1^\infty z^{-\alpha} dz = \alpha \cdot \frac{1}{\alpha-1} = \frac{\alpha}{\alpha-1} < \infty$ (here $\alpha > 1$).

The requested divergence. On the event $\{|Z_1| > a\}$ we have $Z_1^2 > a^2$, so $\mathbb{E}[Z_1^2; |Z_1| > a] \geq a^2 \mathbb{P}(|Z_1| > a)$. Take $a = \varepsilon\sqrt{n}$ and multiply by n :

$$n \mathbb{E}[Z_1^2; |Z_1| > \varepsilon\sqrt{n}] \geq n (\varepsilon\sqrt{n})^2 \mathbb{P}(|Z_1| > \varepsilon\sqrt{n}).$$

Now compute the two factors. First $(\varepsilon\sqrt{n})^2 = \varepsilon^2 n$. Second, since $\mathbb{P}(|Z_1| > z) = z^{-\alpha}$,

$$\mathbb{P}(|Z_1| > \varepsilon\sqrt{n}) = (\varepsilon\sqrt{n})^{-\alpha} = \varepsilon^{-\alpha} (\sqrt{n})^{-\alpha} = \varepsilon^{-\alpha} n^{-\alpha/2}.$$

Substituting,

$$n \mathbb{E}[Z_1^2; |Z_1| > \varepsilon\sqrt{n}] \geq n \cdot \varepsilon^2 n \cdot \varepsilon^{-\alpha} n^{-\alpha/2} = \varepsilon^{2-\alpha} n^{2-\alpha/2} \xrightarrow{n \rightarrow \infty} \infty,$$

since the exponent $2 - \frac{\alpha}{2} > 0$. (In fact $\mathbb{E}[Z_1^2; |Z_1| > a] = \alpha \int_a^\infty z^{1-\alpha} dz = +\infty$ for every finite a , so the quantity is literally $+\infty$; the bound above exhibits the divergence rate.) For the natural array $Z_{nm} = Z_m/\sqrt{n}$ the Lindeberg sum is exactly $\mathbb{E}[Z_1^2; |Z_1| > c\sqrt{n}] = +\infty$, so Lindeberg’s condition cannot hold.

No non-degenerate normal limit. Suppose, for contradiction, $S_n \xrightarrow{d} N(0, \sigma^2)$ for some $\sigma^2 \in [0, \infty)$.

Why it must fail (*necessary condition*). A Gaussian limit has no “jump” part, which forces the individual-term tail mass $n\mathbb{P}(|Z_1| > \varepsilon\sqrt{n}) \rightarrow 0$. Here instead

$$n\mathbb{P}(|Z_1| > \varepsilon\sqrt{n}) = n \cdot \varepsilon^{-\alpha} n^{-\alpha/2} = \varepsilon^{-\alpha} n^{1-\alpha/2} \rightarrow \infty,$$

because $1 - \frac{\alpha}{2} > 0$. This already signals impossibility; we now make it rigorous.

Rigorous contradiction (non-tightness). Let $S_k^\Sigma = \sum_{i=1}^k Z_i$ (unscaled partial sums). Since $Z_k = S_k^\Sigma - S_{k-1}^\Sigma$,

$$|Z_k| \leq |S_k^\Sigma| + |S_{k-1}^\Sigma| \leq 2 \max_{j \leq n} |S_j^\Sigma|, \quad \text{so} \quad \{M_n \geq t\} \subseteq \left\{ \max_{j \leq n} |S_j^\Sigma| \geq t/2 \right\}.$$

By Lévy’s maximal inequality for symmetric independent variables, $\mathbb{P}(\max_{j \leq n} |S_j^\Sigma| \geq t/2) \leq 2\mathbb{P}(|S_n^\Sigma| \geq t/2)$. Combining,

$$\mathbb{P}(|S_n^\Sigma| \geq t/2) \geq \frac{1}{2} \mathbb{P}(M_n \geq t).$$

Take $t = 2K\sqrt{n}$ (so $t/2 = K\sqrt{n}$ and $|S_n^\Sigma| \geq K\sqrt{n} \iff |S_n| \geq K$):

$$\mathbb{P}(|S_n| \geq K) \geq \frac{1}{2} \mathbb{P}(M_n \geq 2K\sqrt{n}).$$

Compute the right side. By independence,

$$\mathbb{P}(M_n \leq 2K\sqrt{n}) = \mathbb{P}(|Z_1| \leq 2K\sqrt{n})^n = (1 - q_n)^n, \quad q_n := \mathbb{P}(|Z_1| > 2K\sqrt{n}) = (2K)^{-\alpha} n^{-\alpha/2}.$$

Using $1 - q_n \leq e^{-q_n}$,

$$(1 - q_n)^n \leq e^{-nq_n} = \exp\left(- (2K)^{-\alpha} n^{1-\alpha/2}\right) \rightarrow 0,$$

so $\mathbb{P}(M_n \geq 2K\sqrt{n}) \rightarrow 1$ and therefore $\liminf_n \mathbb{P}(|S_n| \geq K) \geq \frac{1}{2}$ for every $K > 0$.

But if $S_n \xrightarrow{d} N(0, \sigma^2)$ with $\sigma^2 \in (0, \infty)$, then at the continuity point K ,

$$\mathbb{P}(|S_n| > K) \rightarrow \mathbb{P}(|N(0, \sigma^2)| > K) = 2(1 - \Phi(K/\sigma)),$$

which tends to 0 as $K \rightarrow \infty$; choose K with $\Phi(K/\sigma) > \frac{3}{4}$, giving a limit $< \frac{1}{2}$ — contradicting $\liminf_n \mathbb{P}(|S_n| \geq K) \geq \frac{1}{2}$. (If $\sigma^2 = 0$ the limit is the point mass at 0 and $\mathbb{P}(|S_n| > K) \rightarrow 0$, an even sharper contradiction.) Hence no normal limit exists; Theorem 1.6 genuinely requires $\mathbb{E}\|Z_1\|^2 < \infty$. ■

■ **Technique** With infinite variance the “average” is dominated by a *single* extreme term. Compute $n\mathbb{P}(|Z_1| > \varepsilon\sqrt{n})$: if it does not vanish, a Gaussian limit is impossible. Lévy’s symmetrisation inequality converts “one big summand” into non-tightness of S_n . (Z_1 lies in the domain of attraction of an α -stable law, normalised by $n^{1/\alpha} \gg \sqrt{n}$.)

4. Glivenko–Cantelli fails without independence

Problem

Let $Z_n = Z_1$ a.s. for all n , with Z_1 having a *continuous* cdf F . Show $F^{(n)}(t) = \mathbf{1}_{\{Z_1 \leq t\}}$ for all n , hence $\sup_{t \in \mathbb{R}} |F^{(n)}(t) - F(t)| = \max\{F(Z_1), 1 - F(Z_1)\} \geq \frac{1}{2}$ a.s., and conclude that Glivenko–Cantelli fails without independence.

■ **Setup** Perfectly dependent sample: all Z_i equal, so the empirical cdf collapses to a single jump. Evaluate the sup-distance to the continuous F on the two regions $t < Z_1$ and $t \geq Z_1$.

■ **Tools used** Definition of $F^{(n)}$; monotonicity and continuity (left-limits) of F ; the elementary bound $\max\{a, 1 - a\} \geq \frac{1}{2}$; (aside) Prop. 1.3: $F(Z_1) \sim \text{Unif}(0, 1)$.

■ Solution

Since $Z_i = Z_1$ for every i , each indicator equals $\mathbf{1}_{\{Z_1 \leq t\}}$, so

$$F^{(n)}(t) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{Z_i \leq t\}} = \frac{1}{n} \cdot n \mathbf{1}_{\{Z_1 \leq t\}} = \mathbf{1}_{\{Z_1 \leq t\}} \quad (\forall n),$$

a single step of height 1 located at $t = Z_1$ (independent of n). Now split the supremum.

Region $t < Z_1$. Here $Z_1 > t$, so $\mathbf{1}_{\{Z_1 \leq t\}} = 0$ and $|F^{(n)}(t) - F(t)| = |0 - F(t)| = F(t)$. As F is nondecreasing, its supremum over this region is the left limit at Z_1 ; continuity gives

$$\sup_{t < Z_1} F(t) = \lim_{t \uparrow Z_1} F(t) = F(Z_1^-) = F(Z_1).$$

Region $t \geq Z_1$. Here $\mathbf{1}_{\{Z_1 \leq t\}} = 1$, and since $F \leq 1$, $|F^{(n)}(t) - F(t)| = |1 - F(t)| = 1 - F(t)$. This is nonincreasing in t , so its supremum over $t \geq Z_1$ is attained at $t = Z_1$:

$$\sup_{t \geq Z_1} (1 - F(t)) = 1 - F(Z_1).$$

Combine. Taking the larger of the two regional suprema,

$$\sup_{t \in \mathbb{R}} |F^{(n)}(t) - F(t)| = \max\{F(Z_1), 1 - F(Z_1)\}.$$

For any $a \in [0, 1]$, since $a + (1 - a) = 1$ the larger of the two is at least their average:

$$\max\{a, 1 - a\} \geq \frac{a + (1 - a)}{2} = \frac{1}{2}.$$

Hence $\sup_t |F^{(n)}(t) - F(t)| \geq \frac{1}{2}$ a.s. This value is constant in n and never $\rightarrow 0$, so Theorem 1.8 fails. (As F is continuous, $F(Z_1) \sim \text{Unif}(0, 1)$ by Prop. 1.3, so the sup-distance equals $\max\{U, 1 - U\} \in [\frac{1}{2}, 1]$.) ■

■ **Technique** A counterexample by *maximal dependence*. Collapsing the sample to one point makes $F^{(n)}$ a single jump; its sup-distance to a continuous F is $\max\{F(Z_1), 1 - F(Z_1)\} \geq \frac{1}{2}$. This isolates exactly which hypothesis (independence, through the SLLN behind G-C) is essential.

5. Why Theorem 1.9 needs a new proof

Problem

Explain why the extended Glivenko–Cantelli Theorem 1.9 cannot be proven by combining Theorem 1.7 (Lindeberg CLT) with Lemma 1.5 (abstract Glivenko–Cantelli).

■ **Setup** Theorem 1.9: for a triangular array (Z_{nm}) with arbitrary cdfs F_{nm} and empirical cdf $F^{(n)}$, one has $\sup_t |F^{(n)}(t) - \bar{F}_n(t)| \xrightarrow{a.s.} 0$, where $\bar{F}_n = \frac{1}{n} \sum_m F_{nm}$ is a *moving* (n -dependent) centering.

■ **Tools used** The precise hypotheses of Lemma 1.5 (a *fixed* bounded, non-decreasing, càdlàg limit with pointwise a.s. convergence) and of Theorem 1.7 (finite-dimensional convergence *in distribution*).

■ Solution

1. **Lemma 1.5 requires a *fixed* limit.** Its hypothesis is a single bounded, non-decreasing, càdlàg function T , together with pointwise convergence $T^{(n)}(x) \xrightarrow{a.s.} T(x)$ and $T^{(n)}(x-) \xrightarrow{a.s.} T(x-)$ to *that* T . In Theorem 1.9 the natural centering $\bar{F}_n$ *changes with* n , so there is no fixed T to feed into the lemma: the statement “ $F^{(n)}(x) \rightarrow \bar{F}_n(x)$ ” compares two moving objects and is not convergence to a fixed limit. Moreover the lemma’s proof pins the (finitely many relevant) jumps of the fixed

T on a grid and controls oscillations between them — impossible when the target drifts with n . A further obstruction: the rows have *different* distributions F_{nm} , so the classical iid SLLN does not even supply a.s. pointwise convergence of $F^{(n)}(x)$ to a fixed function.

2. Theorem 1.7 gives the wrong mode and scale. Lindeberg’s CLT yields only *convergence in distribution* of $\sqrt{n}(F^{(n)}(t) - \bar{F}_n(t))$ at finitely many fixed t . That is: (i) in law, not almost surely; (ii) finite-dimensional, not uniform in t ; (iii) it describes fluctuations of order $n^{-1/2}$, not the vanishing of the sup-distance. So it cannot provide the a.s. pointwise input Lemma 1.5 would require, nor the uniformity in t .

Thus the two tools are mismatched, and a genuinely different, array-adapted argument is needed (partition/bracketing handling the moving centering and the supremum simultaneously; cf. Shorack–Wellner, Thm 3.2.1). ■

■ **Technique** Audit hypotheses literally. Two recurring failure modes in “why can’t we just combine...”: (a) a lemma needs a *fixed* object where the problem supplies a *moving* one; (b) a theorem delivers the *wrong convergence mode* (weak vs. a.s.) or *wrong scale*.

6. Converging-together (asymptotic equivalence)

Problem

Let $(Z_n), (W_n)$ be $\mathbb{R}^p$ -valued. Show that if $\|Z_n - W_n\| = o_P(1)$ and $Z_n \xrightarrow{d} Z$, then $W_n \xrightarrow{d} Z$.
Hint: use the definition of convergence in distribution.

■ **Setup** By Definition 1.2 it suffices to prove $\mathbb{E}f(W_n) \rightarrow \mathbb{E}f(Z)$ for every bounded Lipschitz f ; fix such an f with $|f| \leq B$ and Lipschitz constant L (i.e. $|f(x) - f(y)| \leq L\|x - y\|$).

■ **Tools used** Bounded-Lipschitz characterisation of $\xrightarrow{d}$; the triangle inequality; splitting an expectation over a good event $\{\|W_n - Z_n\| \leq \delta\}$ and its rare complement.

■ Solution

Insert the intermediate term $\mathbb{E}f(Z_n)$ and use the triangle inequality:

$$|\mathbb{E}f(W_n) - \mathbb{E}f(Z)| \leq \underbrace{|\mathbb{E}f(W_n) - \mathbb{E}f(Z_n)|}_{\text{(I)}} + \underbrace{|\mathbb{E}f(Z_n) - \mathbb{E}f(Z)|}_{\text{(II)}}.$$

Term (II) $\rightarrow 0$ directly from $Z_n \xrightarrow{d} Z$ (definition, since f is bounded continuous).

For (I), bound $|\mathbb{E}f(W_n) - \mathbb{E}f(Z_n)| \leq \mathbb{E}|f(W_n) - f(Z_n)|$ and split the expectation over $\{\|W_n - Z_n\| \leq \delta\}$ and its complement:

$$\mathbb{E}|f(W_n) - f(Z_n)| = \mathbb{E}[|f(W_n) - f(Z_n)|; \|W_n - Z_n\| \leq \delta] + \mathbb{E}[|f(W_n) - f(Z_n)|; \|W_n - Z_n\| > \delta].$$

On $\{\|W_n - Z_n\| \leq \delta\}$, the Lipschitz bound gives $|f(W_n) - f(Z_n)| \leq L\|W_n - Z_n\| \leq L\delta$, so that piece is $\leq L\delta$. On the complement, $|f(W_n) - f(Z_n)| \leq |f(W_n)| + |f(Z_n)| \leq 2B$, so that piece is $\leq 2B\mathbb{P}(\|W_n - Z_n\| > \delta)$. Hence

$$\mathbb{E}|f(W_n) - f(Z_n)| \leq L\delta + 2B\mathbb{P}(\|W_n - Z_n\| > \delta).$$

Since $\|Z_n - W_n\| = o_P(1)$, the probability $\rightarrow 0$, so $\limsup_n \text{(I)} \leq L\delta$. As $\delta > 0$ is arbitrary, (I) $\rightarrow 0$. Combining, $\mathbb{E}f(W_n) \rightarrow \mathbb{E}f(Z)$ for all bounded Lipschitz f , i.e. $W_n \xrightarrow{d} Z$. ■

Alternative (Slutsky + CMT). From $Z_n \xrightarrow{d} Z$ and $W_n - Z_n \xrightarrow{P} 0 \xrightarrow{d} 0$, Slutsky (Lemma 1.3) gives $(Z_n, W_n - Z_n) \xrightarrow{d} (Z, 0)$; the continuous map $(a, b) \mapsto a + b$ (CMT, Thm 1.4) yields $W_n = Z_n + (W_n - Z_n) \xrightarrow{d} Z + 0 = Z$.

■ **Technique** The workhorse lemma of asymptotics (cited throughout the chapter as “Exercise 6”). Master the *two-term split + Lipschitz/bounded dichotomy*: bound $|f(W_n) - f(Z_n)|$ by $L\delta$ on a good event and by $2B$ on a rare event.

7. Details of Example 1.5 (the o_P/O_P transfer rule)

Problem

Go through the details of Example 1.5 — if $Z_n = O_P(a_n)$ and $W_n = o_P(\|Z_n\|)$ then $W_n = o_P(a_n)$ — and discuss its practical implications.

■ **Setup** Definitions ($a_n > 0$ deterministic): $Z_n = O_P(a_n)$: $\lim_{M \rightarrow \infty} \limsup_n \mathbb{P}(\|Z_n\|/a_n > M) = 0$; $W_n = o_P(\|Z_n\|)$: $\forall \varepsilon > 0, \mathbb{P}(\|W_n\|/\|Z_n\| > \varepsilon) \rightarrow 0$; goal $W_n = o_P(a_n)$: $\forall \varepsilon > 0, \mathbb{P}(\|W_n\|/a_n > \varepsilon) \rightarrow 0$.

■ **Tools used** One pointwise set-inclusion; the two-event bound $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$; careful quantifier order (fix the auxiliary constant from O_P before invoking o_P).

Solution

Fix $\varepsilon, \delta > 0$. Introduce a free parameter $\varepsilon_1 > 0$ (chosen below). The key algebraic observation is the factorisation $\frac{\|W_n\|}{a_n} = \frac{\|W_n\|}{\|Z_n\|} \cdot \frac{\|Z_n\|}{a_n}$, which yields the pointwise inclusion

$$\left\{ \frac{\|W_n\|}{\|Z_n\|} < \varepsilon_1 \right\} \cap \left\{ \frac{\|Z_n\|}{a_n} < \frac{\varepsilon}{\varepsilon_1} \right\} \subseteq \left\{ \frac{\|W_n\|}{a_n} < \varepsilon \right\},$$

because on the intersection $\frac{\|W_n\|}{a_n} < \varepsilon_1 \cdot \frac{\varepsilon}{\varepsilon_1} = \varepsilon$. Taking probabilities and using $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$ (which follows from $\mathbb{P}(A \cup B) \leq 1$ and inclusion–exclusion),

$$\mathbb{P}\left(\frac{\|W_n\|}{a_n} < \varepsilon\right) \geq \mathbb{P}\left(\frac{\|W_n\|}{\|Z_n\|} < \varepsilon_1\right) + \mathbb{P}\left(\frac{\|Z_n\|}{a_n} < \frac{\varepsilon}{\varepsilon_1}\right) - 1. \quad (\dagger)$$

Order of choices (the crux). First use O_P : it provides M_0 with $\limsup_n \mathbb{P}(\|Z_n\|/a_n > M_0) < \delta/2$; fix $\varepsilon_1 := \varepsilon/M_0$, so that $\varepsilon/\varepsilon_1 = M_0$. Then there is L_1 with, for $n \geq L_1$,

$$\mathbb{P}(\|Z_n\|/a_n < \varepsilon/\varepsilon_1) = \mathbb{P}(\|Z_n\|/a_n < M_0) \geq 1 - \mathbb{P}(\|Z_n\|/a_n > M_0) > 1 - \frac{\delta}{2}.$$

Now, with ε_1 frozen, use o_P : there is L_2 with $\mathbb{P}(\|W_n\|/\|Z_n\| < \varepsilon_1) > 1 - \frac{\delta}{2}$ for $n \geq L_2$. Plugging both into $(\dagger)$, for $n \geq L := \max(L_1, L_2)$,

$$\mathbb{P}(\|W_n\|/a_n < \varepsilon) \geq (1 - \frac{\delta}{2}) + (1 - \frac{\delta}{2}) - 1 = 1 - \delta,$$

i.e. $\mathbb{P}(\|W_n\|/a_n \geq \varepsilon) \leq \delta$ for all large n . Since ε, δ were arbitrary, $W_n = o_P(a_n)$. ■ *(Ratios by $\|Z_n\|$ are read on $\{Z_n \neq 0\}$; if $\mathbb{P}(Z_n = 0) > 0$ then $W_n = o_P(\|Z_n\|)$ forces $W_n = 0$ there, which only helps.)*

Practical implications. This is the *transfer rule* of the o_P/O_P calculus: a remainder small relative to $\|Z_n\|$ is automatically small at the absolute rate a_n , once the rate of Z_n is known. Its canonical use is inside the delta method (Thm 1.10): a Taylor remainder is naturally $o_P(\|T_n - c\|)$; a CLT gives $T_n - c = O_P(a_n^{-1})$; the rule turns the remainder into $o_P(a_n^{-1})$, which vanishes after multiplying by a_n . It licenses the shorthand “ $o_P(\|Z_n\|) = o_P(\text{rate of } Z_n)$ ” and shows why pinning the rate $O_P(a_n)$ of an estimator is always the crucial first step.

■ **Technique** Stochastic-order arithmetic via one set-inclusion + $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$. The only subtlety is *quantifier order*: extract M_0 (hence fix ε_1) from the uniform-in- n O_P tail before invoking the per- ε o_P statement.

8. Multivariate CLT for the empirical cdf

Problem

Prove the multidimensional version of (1.4): for $-\infty < t_1 < \dots < t_M < \infty$,

$$\sqrt{n}(F^{(n)}(t_1) - F(t_1), \dots, F^{(n)}(t_M) - F(t_M)) \xrightarrow{d} N(0, \Sigma), \quad \Sigma_{ij} = F(t_i \wedge t_j) - F(t_i)F(t_j).$$

■ **Setup** Package the M indicators into one iid random vector, reducing the whole problem to computing a mean vector and a covariance matrix.

■ **Tools used** Laplace's multivariate CLT (Thm 1.6); the identity $\mathbf{1}_A \mathbf{1}_B = \mathbf{1}_{A \cap B}$; (mechanism) the Cramér–Wold device.

■ Solution

Define the iid $\mathbb{R}^M$ -valued vectors

$$\xi_i = (\mathbf{1}_{\{Z_i \leq t_1\}}, \dots, \mathbf{1}_{\{Z_i \leq t_M\}})^\top, \quad i = 1, \dots, n,$$

which are iid because the Z_i are. Averaging coordinatewise,

$$\frac{1}{n} \sum_{i=1}^n \xi_i = \left(\frac{1}{n} \sum_i \mathbf{1}_{\{Z_i \leq t_1\}}, \dots \right)^\top = (F^{(n)}(t_1), \dots, F^{(n)}(t_M))^\top =: \bar{\xi}_n.$$

Mean. Coordinate k of $\mathbb{E}\xi_i$ is $\mathbb{E}\mathbf{1}_{\{Z_i \leq t_k\}} = \mathbb{P}(Z_i \leq t_k) = F(t_k)$, so $\mathbb{E}\xi_i = (F(t_1), \dots, F(t_M))^\top =: \mu$.

Covariance. For entry (k, l) we need $\text{Cov}(\mathbf{1}_{\{Z_1 \leq t_k\}}, \mathbf{1}_{\{Z_1 \leq t_l\}})$. The product of indicators is the indicator of the intersection, and since the t 's are ordered, $\{Z_1 \leq t_k\} \cap \{Z_1 \leq t_l\} = \{Z_1 \leq \min(t_k, t_l)\} = \{Z_1 \leq t_k \wedge t_l\}$:

$$\mathbb{E}[\mathbf{1}_{\{Z_1 \leq t_k\}} \mathbf{1}_{\{Z_1 \leq t_l\}}] = \mathbb{E}\mathbf{1}_{\{Z_1 \leq t_k \wedge t_l\}} = \mathbb{P}(Z_1 \leq t_k \wedge t_l) = F(t_k \wedge t_l).$$

Hence

$$\text{Cov}(\mathbf{1}_{\{Z_1 \leq t_k\}}, \mathbf{1}_{\{Z_1 \leq t_l\}}) = \mathbb{E}[\mathbf{1} \mathbf{1}] - \mathbb{E}[\mathbf{1}] \mathbb{E}[\mathbf{1}] = F(t_k \wedge t_l) - F(t_k)F(t_l) = \Sigma_{kl},$$

so $\text{Var}(\xi_1) = \Sigma$. The second moment is finite: $\mathbb{E}\|\xi_1\|^2 = \sum_{k=1}^M \mathbb{E}\mathbf{1}_{\{Z_1 \leq t_k\}}^2 = \sum_{k=1}^M F(t_k) \leq M < \infty$ (indicators are 0/1, so $\mathbf{1}^2 = \mathbf{1}$).

Apply Laplace's CLT (Thm 1.6). The ξ_i are iid with $\mathbb{E}\|\xi_1\|^2 < \infty$, hence

$$\sqrt{n}(\bar{\xi}_n - \mu) \xrightarrow{d} N(0, \text{Var}(\xi_1)) = N(0, \Sigma),$$

which is exactly the claimed statement. ■

Mechanism (Cramér–Wold). Equivalently, for any $a \in \mathbb{R}^M$, $a^\top \xi_i = \sum_k a_k \mathbf{1}_{\{Z_i \leq t_k\}}$ is a real iid sequence with variance $a^\top \Sigma a < \infty$; the univariate CLT gives $\sqrt{n} a^\top (\bar{\xi}_n - \mu) \xrightarrow{d} N(0, a^\top \Sigma a)$, and Cramér–Wold assembles the vector statement. *Checks:* the diagonal is $\Sigma_{kk} = F(t_k \wedge t_k) - F(t_k)^2 = F(t_k)(1 - F(t_k))$ (Bernoulli variance); off-diagonals are ≥ 0 (positive dependence). These are the finite-dimensional covariances of $B \circ F$ (Thm 1.22, Brownian-bridge structure).

■ **Technique** Vectorise, then hit with Laplace's CLT. For empirical-cdf problems, pack the indicators at $t_1, \dots, t_M$ into one iid vector ξ_i ; the whole problem reduces to $\mathbb{E}\xi_1$ and $\text{Var}(\xi_1)$, where $\mathbf{1}_A \mathbf{1}_B = \mathbf{1}_{A \cap B}$ produces the $t_i \wedge t_j$ covariance. This is the finite-dimensional half of the functional CLT.

Part II — General techniques, purpose, and exam relevance

A. A catalogue of transferable proof techniques

These recur far beyond the eight problems; each is a reusable “move”.

1. **Symmetrisation.** Apply a one-sided result to $-\gamma$, or use distributional symmetry to reduce a sum to a single (max) term. (*Ex 1; Ex 3 via Lévy.*)
2. **Single-variable reparametrisation.** When two quantities are tied by an equation, substitute to kill radicals and cancel common factors. (*Ex 1: $s = \sqrt{2t}$, then $x = \kappa s$.*)
3. **Bound below by one term.** Drop nonnegative summands ($\sum_i Z_i \geq Z_n$) or keep only the maximum; a statement about a whole sum becomes a one-term tail. (*Ex 2, Ex 3.*)
4. **Truncation + tail integrals.** Split $\mathbb{E}[X^2] = \mathbb{E}[X^2; |X| \leq a] + \mathbb{E}[X^2; |X| > a]$; evaluate Pareto integrals $\int z^p dz$ and read convergence off the sign of the exponent. (*Ex 2, Ex 3.*)
5. **Borel–Cantelli.** Convert summability of $\sum \mathbb{P}(A_n)$ into a.s. statements; remember both directions, and that “i.o.” needs *independence + divergence*. (*Ex 2.*)
6. **Concentration $\rightarrow$ two-sided $\rightarrow$ rate.** Union-bound a one-sided exponential inequality; invert the threshold to expose an explicit rate; identify the sub-Gaussian/sub-exponential regimes. (*Ex 1.*)
7. **Triangle inequality with an inserted pivot.** $|\mathbb{E}f(W_n) - \mathbb{E}f(Z)| \leq |\mathbb{E}f(W_n) - \mathbb{E}f(Z_n)| + |\mathbb{E}f(Z_n) - \mathbb{E}f(Z)|$. (*Ex 6.*)
8. **Good-event / rare-event split.** Bound the integrand by a small Lipschitz term on $\{\|\cdot\| \leq \delta\}$ and by a crude constant on the rare complement. (*Ex 6.*)
9. **Set-inclusion + $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$, with quantifier discipline.** The backbone of o_P/O_P manipulations; fix constants from the *uniform* (O_P) statement before using the *per- ε* (o_P) one. (*Ex 7.*)
10. **Partition-then-sup.** Control a supremum by pointwise values on a finite grid plus the oscillation between grid points. (*Ex 4; abstract Glivenko–Cantelli, Lemma 1.5; Dini, Thm 1.15.*)
11. **Vectorise + Cramér–Wold / Laplace.** Reduce a multivariate limit to computing a mean and covariance (or to a family of 1-D limits). (*Ex 8.*)
12. **Hypothesis audit.** To test whether “ $A + B \Rightarrow C$ ”, check each hypothesis literally: fixed vs. moving limit, weak vs. a.s., pointwise vs. uniform, and the scaling. (*Ex 5.*)

B. What each exercise is meant to illustrate

Ex. 1 — Non-asymptotic control. Asymptotic (CLT) statements have finite- n analogues. Bernstein’s inequality lets you *choose* a simulation/Monte-Carlo size n for a target accuracy ε and confidence $1 - \eta$ (Remark 1.2). Illustrates the concentration toolkit and the sub-Gaussian/sub-exponential crossover.

Ex. 2 & 3 — Necessity of moment hypotheses. The (strong) law needs a finite mean; the CLT needs a finite variance. With heavy tails — *exactly the regime of large insurance claims* — these fail: the sample mean is unstable (Ex 2), and normalised sums converge to *stable*, not normal, laws (Ex 3). Purpose: locate the boundary of validity of Theorems 1.2 and 1.6, and motivate the domain-of-attraction / extreme-value viewpoint used later in the book.

Ex. 4 & 5 — Independence, and pointwise vs. uniform. Ex 4 shows Glivenko–Cantelli genuinely *uses* independence (a fully dependent sample keeps the sup-distance $\geq \frac{1}{2}$). Ex 5 shows why array uniform laws need dedicated machinery: a CLT plus an abstract G–C lemma cannot yield an *a.s. uniform* statement against a *moving* centering. Purpose: sharpen intuition for what actually makes empirical-process results run.

Ex. 6 & 7 — The everyday calculus of asymptotics. These are the lemmas used silently in *every* delta-method and Slutsky argument: asymptotic equivalence / “converging together” (Ex 6) and the o_P/O_P transfer rule (Ex 7). Purpose: supply the rigorous underpinning so later hand-waving is safe — e.g. replacing an unknown σ by a consistent $\hat{\sigma}_n$, or discarding a Taylor remainder.

Ex. 8 — The skeleton of the empirical process. Establishes multivariate asymptotic normality of the empirical cdf at finitely many thresholds, with Brownian-bridge covariance $F(s \wedge t) - F(s)F(t)$. These are the finite-dimensional distributions that, together with tightness, give the functional CLT (Thm 1.22). Purpose: bridge from the classical CLT to the process-level results behind nonparametric quantile/VaR, mean-excess, cumulative-hazard, and copula inference.

Insurance thread. Claim severities are heavy-tailed, so Ex 2–3 are a warning label on naive averaging and Gaussian approximations. The exploratory functionals of §1.3 (empirical variance, mean-excess, cumulative hazard, quantiles/VaR) are handled by exactly the finite-dimensional CLT of Ex 8 *plus* the delta method, whose remainder control is Ex 7; their uncertainty bands are the finite-sample use of Ex 1’s philosophy.

C. Exam trigger guide — “if you see . . . , reach for . . .”

- “Show the (S)LLN fails for this heavy-tailed law.” $\Rightarrow$ moments via the tail integral; Borel–Cantelli with $\{Z_n > Mn\}$; positivity to reach the average. (Ex 2.)
- “Show the CLT fails / the limit is not normal when $\mathbb{E}X^2 = \infty$.” $\Rightarrow$ compute $n\mathbb{P}(|Z_1| > \varepsilon\sqrt{n})$; if it does not vanish, no normal limit; make it rigorous with Lévy’s inequality / the max term. (Ex 3.)
- “Give a non-asymptotic error bound / pick n so the error is $< \varepsilon$ w.p. $\geq 1 - \eta$.” $\Rightarrow$ two-sided Bernstein, then solve the exponent for n . (Ex 1.)
- “Prove $W_n \xrightarrow{d} Z$ knowing $W_n \approx Z_n$ ”, or “may I replace $\hat{\sigma}_n$ by σ / $\hat{\theta}_n$ by θ ?” $\Rightarrow$ converging-together lemma / Slutsky. (Ex 6.)
- “Show the remainder is asymptotically negligible” (any delta-method step). $\Rightarrow$ the o_P/O_P transfer rule; first pin the rate $O_P(a_n^{-1})$. (Ex 7.)
- “Find the joint/limit distribution of the empirical cdf (or a functional of it) at several thresholds.” $\Rightarrow$ vectorise the indicators + Laplace CLT, then delta method for the functional (variance, mean-excess, hazard, quantiles). (Ex 8, then §1.3.)
- “Why is independence / this hypothesis needed?” or “why can’t we prove X by combining Y and Z ?” $\Rightarrow$ break the hypothesis with a counterexample (Ex 4) or run a hypothesis audit (Ex 5).
- “Prove uniform (not just pointwise) convergence of a monotone / càdlàg object.” $\Rightarrow$ partition-then-sup: grid values + oscillation control (Glivenko–Cantelli / Dini). (Ex 4 technique, Lemma 1.5.)

Exam-preparation cheat sheet

Ex.	Core method	One-line trigger
1	Bernstein + symmetrisation; threshold→rate	set $\varepsilon = \kappa t + \sqrt{2t}$; sub $s = \sqrt{2t}$, $x = \kappa s$
2	2nd Borel–Cantelli (independence)	$\sum \mathbb{P}(Z_n > Mn) = \infty \Rightarrow \text{SLLN fails}$
3	tail necessary condition + Lévy inequality	$n\mathbb{P}(Z_1 > \varepsilon\sqrt{n}) \not\rightarrow 0 \Rightarrow \text{no normal}$
4	explicit counterexample (max dependence)	$\sup = \max\{F(Z_1), 1 - F(Z_1)\} \geq \frac{1}{2}$
5	hypothesis audit (fixed vs. moving; mode/scale)	Lemma 1.5 needs <i>fixed</i> T ; CLT is weak
6	bounded-Lipschitz split (or Slutsky + CMT)	$\mathbb{E} f(W_n) - f(Z_n) \leq L\delta + 2B \mathbb{P}(\dots)$
7	o_P/O_P transfer via one set-inclusion	fix ε_1 from O_P <i>before</i> using o_P
8	vectorise indicators + Laplace CLT	$\xi_i = (\mathbf{1}_{Z_i \leq t_k})_k$; $\mathbf{1}_A \mathbf{1}_B = \mathbf{1}_{A \cap B}$

Recurring skills to drill: the tail-integral identity $\mathbb{E}Y = \int_0^\infty \mathbb{P}(Y > z) dz$; both directions of Borel–Cantelli; Cramér–Wold to lift 1-D CLTs; the bounded-Lipschitz characterisation of $\xrightarrow{d}$; and the o_P/O_P algebra behind every delta-method argument.